

BANARAS HINDU UNIVERSITY

B.A./B.Sc. (Hons.) Semester V Examination, 2022-23

Paper No. STA/STB-503: Programming with C

Time: 3 Hours

Maximum Marks: 70

*Instructions: Answer any **five** questions. All questions carry equal marks. Symbols have their usual meanings.*

Question 1.

- (a) Explain type conversions in C expressions with examples. Discuss how operator precedence and associativity rules are applied. [7]
- (b) Determine the value of each of the following logical expressions, given `int a = 5, b = 10, c = -1`:
- (i) `a > b && a < c`
 - (ii) `a < b && a > c`
 - (iii) `a == c || b > a`
 - (iv) `b > 15 && c < 0 || a > 0`

[7]

Question 2.

- (a) Describe the different forms of the `if-else` statement. How do they differ? Write a C program to calculate the mean of 10 numbers entered by the user. [7]
- (b) Write the output of the following program:

```
#include <stdio.h>
int main() {
    int x = 100;
    printf("%d\n", 10 + x++ - 4);
    printf("%d\n", 10 + --x + x);
    return 0;
}
```

[7]

Question 3.

- (a) Define one-dimensional and two-dimensional arrays in C. Explain how a one-dimensional array is declared and initialized. Write a C program to find the transpose of a matrix. [7]

- (b) What is the output of the following program?

```
#include <stdio.h>
int main() {
    int m[] = {1, 2, 3, 4, 5};
    int x, y = 0;
    for (x = 0; x < 5; x++)
        y = y + m[x];
    printf("%d", y);
    return 0;
}
```

[7]

Question 4.

- (a) Discuss how string variables are declared and initialized in C. Explain how strings are read from the terminal and written to the screen. Write a C program (without using library string functions) to copy one string into another and count the number of characters copied. [7]
- (b) Given `char string1[10] = "abc", string2[20] = "def", string3[30], string4[40]`, write the output of the following statements:

```
printf("%s", strcpy(string3, string1));
printf("%s", strcat(strcat(strcpy(string4, string1), "or"), string2));
printf("%d %d", strlen(string2) + strlen(string3), strlen(string4));
```

[7]

Question 5.

- (a) Explain the different categories of functions in C with proper examples. Discuss the relevance of storage classes on the scope, visibility, and lifetime of variables. Write a C program that uses a function to sort an array of integers. [7]
- (b) What are user-defined functions? State three advantages of using functions. What is the purpose of a `return` statement? [7]

Question 6.

- (a) The Fibonacci numbers are defined recursively as: $F_1 = 1$, $F_2 = 2$, $F_n = F_{n-1} + F_{n-2}$ for $n > 2$. Write a C program containing a recursive function to generate and print the first 20 Fibonacci numbers. [7]
- (b) What is printed when the following program is executed?

```
#include <stdio.h>
int f(int);
```

```
int main() {  
    printf("%d\n", f(7));  
    return 0;  
}  
int f(int x) {  
    if (x <= 4) return x;  
    return f(x - 1) + f(x - 2);  
}
```

[7]

Question 7. With examples and programs, explain the following concepts in C. Give a typical use case for each: [14]

- (a) Structures
- (b) Array of structures
- (c) Nested structures
- (d) Unions (include similarities and differences with structures, and rules for initializing)